

1 Database Design

A robust and efficient database design serves as the backbone of **Chaldal.com's e-commerce platform**, ensuring seamless data flow across departments—from inventory management and order processing to customer support and delivery tracking. The proposed database system is designed to handle large volumes of transactions securely and efficiently while maintaining scalability to accommodate future business growth.

1.1 Objectives

The primary objective of the database design for **Chaldal Ltd.**, Bangladesh's largest online grocery delivery service, is to develop a **robust, efficient, and scalable data management system** that supports both operational and strategic needs. The design aims to integrate all functional domains—inventory, order processing, payment handling, customer service, and delivery management—into a unified and optimized data ecosystem, thereby ensuring smooth coordination and informed decision-making throughout the organization.

A well-structured and normalized database will not only enhance the performance and reliability of daily operations but also enable **data-driven insights** for long-term business growth.

1.1.1 Specific Objectives

1. **Efficient Data Organization and Storage:** Design a centralized data repository for structured storage and efficient retrieval of information related to customers, products, orders, inventory, and deliveries. This ensures coherence, accessibility, and consistency across all business units.
2. **Ensuring Data Integrity and Consistency:** Apply strong constraints, referential integrity, and validation mechanisms to maintain data accuracy and prevent anomalies, ensuring consistency across related entities.
3. **Data Security and Confidentiality:** Implement comprehensive security policies including data encryption, user authentication, and role-based access control to protect sensitive customer and company information from unauthorized access.
4. **Minimization of Data Redundancy:** Use normalization techniques to eliminate duplicate data, reduce storage overhead, and ensure that updates occur seamlessly without inconsistencies.
5. **Support for Multi-User Environment:** Develop a concurrent database system enabling multiple users—such as administrators, warehouse staff, delivery agents, and customer service personnel—to access and modify data simultaneously without performance degradation.
6. **System Scalability and Extensibility:** Ensure that the database can scale with business growth, accommodating an expanding product catalog, user base, and transaction volume, and adapting to new features or technologies.
7. **Enhanced Transaction Management:** Guarantee reliable transaction handling using ACID (Atomicity, Consistency, Isolation, Durability) properties to maintain data reliability during operations such as order placement, payment processing, and inventory updates.

8. **Comprehensive Reporting and Analytical Capabilities:** Integrate interrelated datasets to facilitate the generation of dynamic analytical reports that support performance monitoring, trend prediction, and informed decision-making.
9. **Operational Efficiency and Process Automation:** Streamline workflows—from order placement to delivery—by automating routine processes, reducing manual intervention, and minimizing human errors.
10. **Improved Data Accessibility and Decision Support:** Provide timely and accurate access to critical information for both technical and managerial users to support efficient decision-making and optimized resource management.
11. **Customer-Centric Data Management:** Maintain comprehensive customer profiles including purchase history, preferences, feedback, and location details to enable personalized recommendations and improve customer satisfaction.
12. **Regulatory Compliance and Data Governance:** Ensure that the database complies with national and international privacy and data governance standards, maintaining transparency in data collection, storage, and processing.
13. **System Reliability and Fault Tolerance:** Design the database to ensure consistent performance with minimal downtime. Incorporate backup and recovery mechanisms to safeguard business continuity in case of data loss or system failure.

By achieving these objectives, the proposed database system will serve as a comprehensive and reliable backbone for Chaldal Ltd. It will enhance operational efficiency, support data-driven decision-making, and improve overall user satisfaction. Furthermore, the design ensures adaptability to evolving market demands and technological advancements, reinforcing Chaldal's position as a leader in Bangladesh's online grocery delivery industry.

1.2 Entity–Relationship Diagram

Overview

The database for **Chaldal Ltd.** integrates multiple entities that represent different operational components of the e-commerce grocery system. It manages information about customers, products, orders, payments, and deliveries in a relational and structured format.

Each entity is linked by key relationships that ensure data integrity and consistency throughout the business process—from product listing and order creation to delivery completion and payment confirmation.

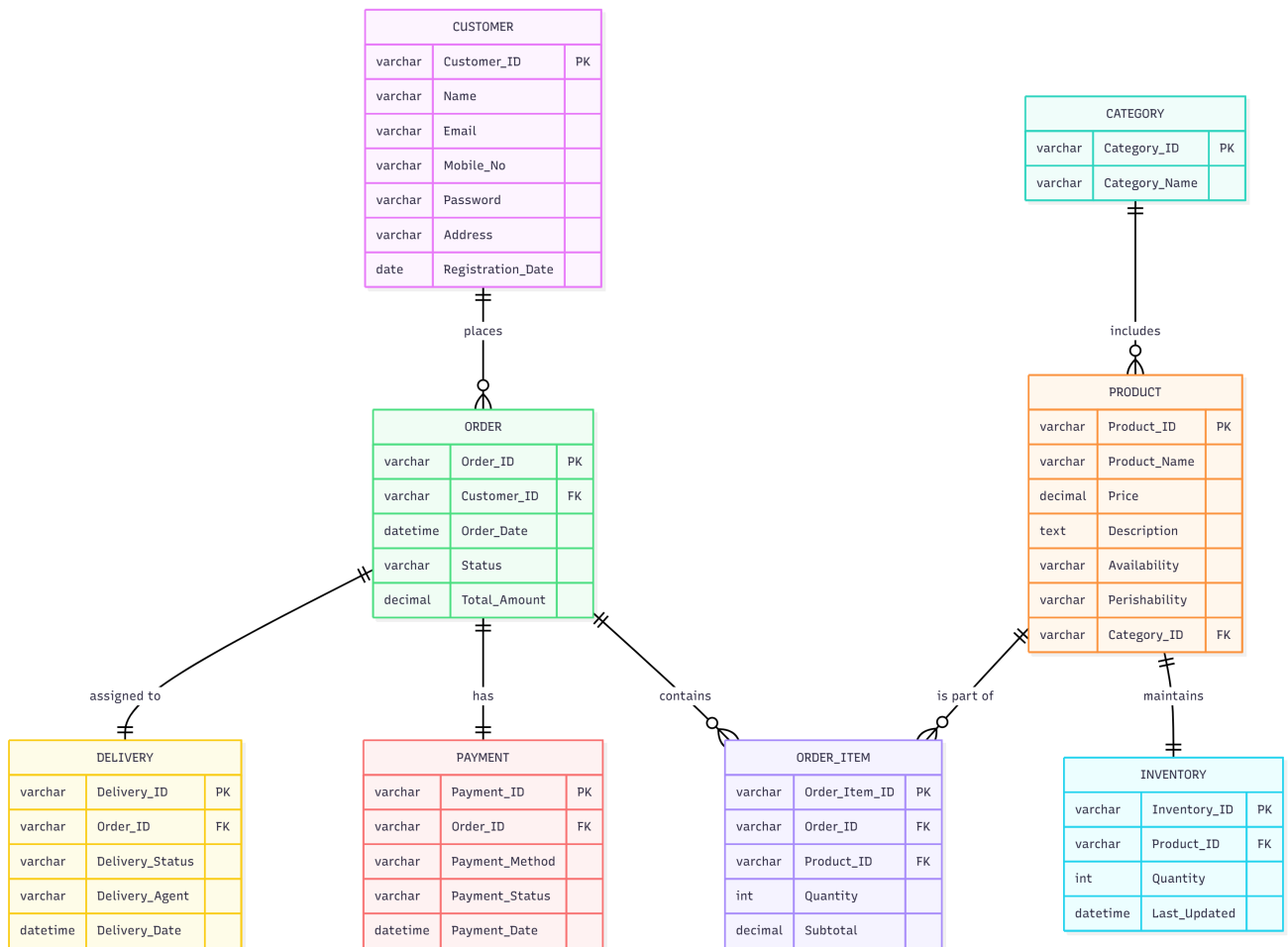


Figure 1: Entity–Relationship Diagram of Chaldal Ltd.

Entities and Their Key Attributes

Customer

- **Customer_ID (PK)**
- Name
- Email
- Mobile_No
- Address
- Password
- Registration_Date

Category

- **Category_ID (PK)**
- Category_Name

Product

- **Product_ID (PK)**
- Product_Name
- Price
- Description
- Perishability
- Availability
- **Category_ID (FK)**

Inventory

- **Inventory_ID (PK)**
- **Product_ID (FK)**
- Quantity
- Last_Updated

Order

- **Order_ID (PK)**
- **Customer_ID (FK)**
- Order_Date
- Status
- Total_Amount

Order_Item

- **Order_Item_ID (PK)**
- **Order_ID (FK)**
- **Product_ID (FK)**
- Quantity
- Subtotal

Payment

- **Payment_ID (PK)**
- **Order_ID (FK)**
- Payment_Method
- Payment_Status
- Payment_Date

Delivery

- **Delivery_ID (PK)**

- **Order_ID (FK)**
- Delivery_Status
- Delivery_Agent
- Delivery_Date

Relationships Among Entities

Relationship	Description	Type
Customer → Order	A customer can have multiple orders, but each order belongs to one customer.	1-to-Many
Category → Product	Each category can include multiple products.	1-to-Many
Product → Inventory	Each product has one inventory record.	1-to-1
Order → Order_Item	Each order can contain multiple products.	1-to-Many
Product → Order_Item	A product can appear in many orders.	1-to-Many
Order → Payment	Each order has one payment transaction associated with it.	1-to-1
Order → Delivery	Each order has one delivery record.	1-to-1

Table 1: Relationships Among Entities in Chaldal Ltd. Database

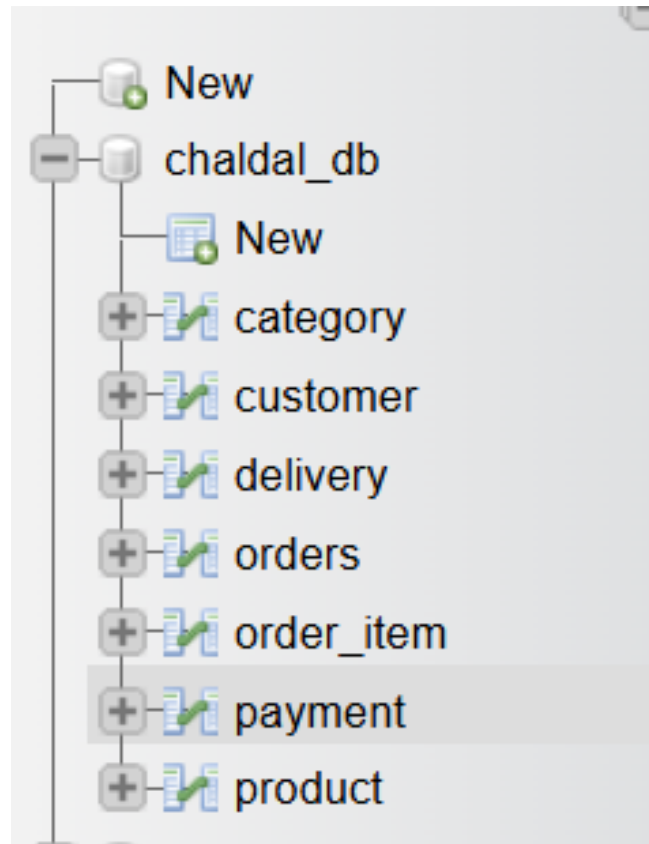


Figure 2: Database of Chaldal

Text-Based Diagram Structure (Simplified Layout)

Description of Entity Relationships

- **Customer–Order Relationship:** Each customer can place multiple orders, while each order must belong to one registered customer.
- **Order–Order_Item Relationship:** Each order can contain multiple order items; each item represents a specific product purchased within that order.
- **Order_Item–Product Relationship:** Each order item corresponds to one product, defining which product is included in the order and at what quantity.
- **Product–Category Relationship:** Products are classified under categories such as *Fruits*, *Vegetables*, *Baby Care*, *Staples*, etc., streamlining product organization and filtering.
- **Product–Inventory Relationship:** Each product maintains an inventory record that tracks its availability, quantity, and update timestamp.
- **Order–Payment Relationship:** Each order has a corresponding payment record that includes method (e.g., cash, card, mobile payment) and status (e.g., pending, paid).
- **Order–Delivery Relationship:** Each order is associated with one delivery transaction that records the delivery status, assigned agent, and completion date.

Visual Representation Guide (Simplified ER Layout)

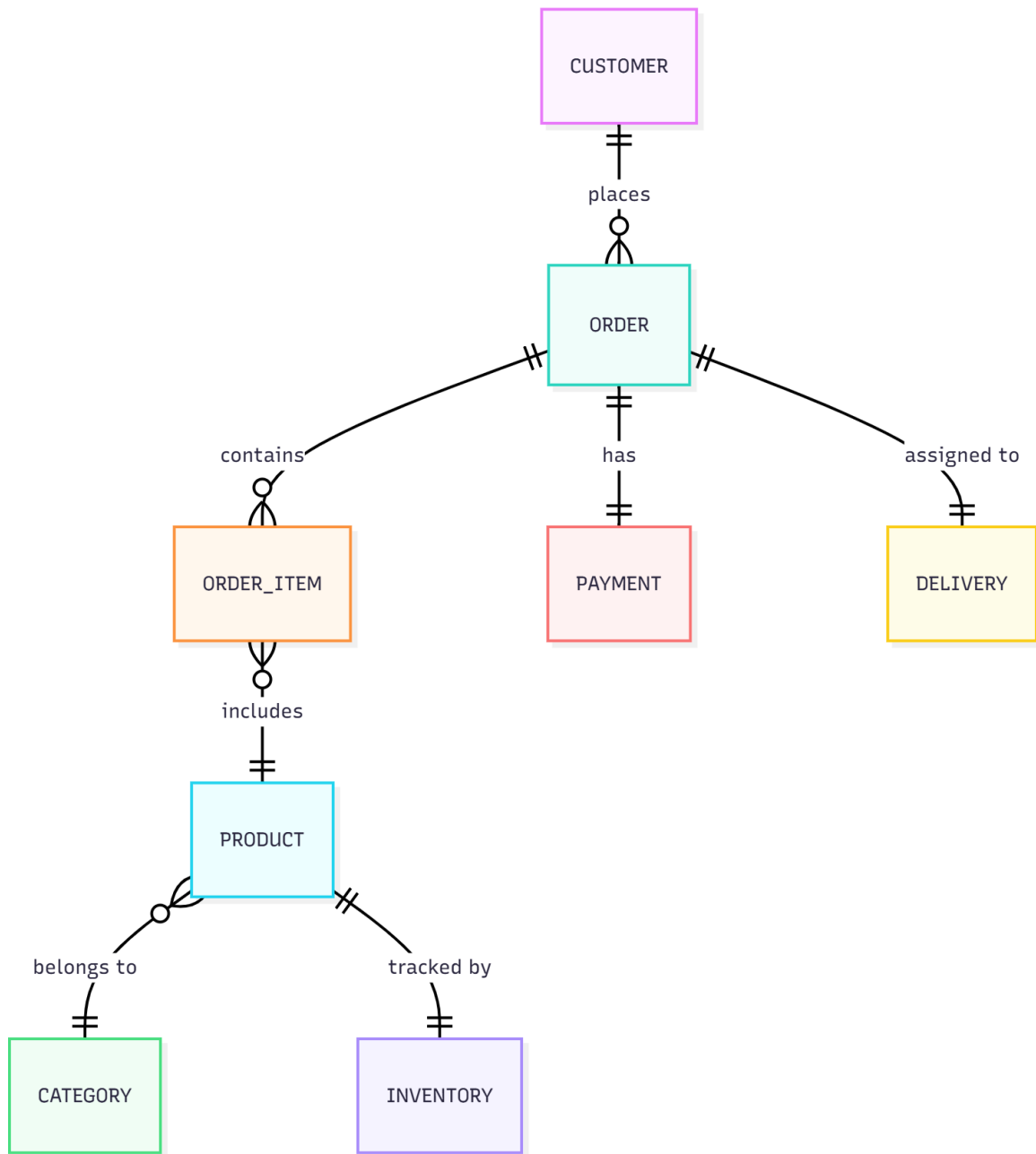


Figure 3: Entity-Relationship Diagram of Chaldal Ltd.

1.3 Database Table Design

4.3.3.1 Customer Table

Field Name	Key Type	Data Type	Size	Description
Customer_ID	Primary Key	VARCHAR	10	Unique identifier for each customer
Name		VARCHAR	50	Full name of the customer
Email		VARCHAR	50	Customer's email address
Mobile_No		VARCHAR	15	Contact number of the customer
Password		VARCHAR	255	Encrypted customer password
Address		VARCHAR	255	Physical or delivery address
Registration_Date		DATE	—	Date when the customer registered

Table 2: Customer Table Design

Customer_ID	Name	Email	Mobile_No	Password	Address	Registration_Date
C001	Rahim Uddin	rahim@gmail.com	01712345678	pass123	Dhaka	2024-02-10
C002	Sabina Akter	sabina@gmail.com	01855678900	sabina22	Chittagong	2024-02-12
C003	Tanvir Hossain	tanvir@outlook.com	01699887766	tan12	Khulna	2024-02-15
C004	Nadia Rahman	nadia.rahman@yahoo.com	01555443377	nadia55	Rajshahi	2024-02-18
C005	Arif Mahmud	arifm@gmail.com	01922334455	arifm99	Sylhet	2024-02-20

Figure 4: Customer Table

4.3.3.2 Category Table

Field Name	Key Type	Data Type	Size	Description
Category_ID	Primary Key	VARCHAR	10	Unique ID for each category
Category_Name		VARCHAR	50	Product category name (e.g., Fruits, Dairy, Household)

Table 3: Category Table Design

Category_ID	Category_Name
CAT01	Fruits
CAT02	Vegetables
CAT03	Dairy
CAT04	Beverages
CAT05	Snacks

Figure 5: Category Table

4.3.3.3 Product Table

Field Name	Key Type	Data Type	Size	Description
Product_ID	Primary Key	VARCHAR	10	Unique product identifier
Product_Name		VARCHAR	100	Name of the product
Price		DECIMAL	(10,2)	Price of the product
Description		TEXT	—	Product description/details
Availability		VARCHAR	5	Product stock status (“Yes/No”)
Perishability		VARCHAR	5	Product perishability (“Yes/No”)
Category_ID	Foreign Key	VARCHAR	10	Linked to Category Table

Table 4: Product Table Design

Product_ID	Product_Name	Price	Description	Availability	Perishability	Category_ID
P001	Apple	120.50	Fresh apples	Yes	Yes	CAT01
P002	Banana	60.00	Organic bananas	Yes	Yes	CAT01
P003	Spinach	40.00	Farm fresh spinach bunch	Yes	Yes	CAT02
P004	Milk 1L	90.00	Fresh pasteurized milk 1 liter	Yes	Yes	CAT03
P005	Yogurt 500g	75.00	Plain yogurt cup 500 grams	Yes	Yes	CAT03
P006	Orange Juice 1L	180.00	Packed orange juice bottle 1L	Yes	No	CAT04
P007	Coca-Cola 1.5L	110.00	Soft drink bottle 1.5L	Yes	No	CAT04
P008	Potato Chips 200g	55.00	Salted potato chips pack	Yes	No	CAT05
P009	Chocolate Bar 50g	80.00	Milk chocolate bar 50 grams	Yes	No	CAT05
P010	Tomato	45.00	Fresh tomato per kg	Yes	Yes	CAT02

Figure 6: Product Table

4.3.3.4 Order Table

Field Name	Key Type	Data Type	Size	Description
Order_ID	Primary Key	VARCHAR	10	Unique order identifier
Customer_ID	Foreign Key	VARCHAR	10	Linked to Customer Table
Order_Date		DATETIME	—	Date and time of order placement
Status		VARCHAR	20	Order status (“Pending”, “Delivered”, “Cancelled”, etc.)
Total_Amount		DECIMAL	(10,2)	Total cost of the order

Table 5: Order Table Design

Order_ID	Customer_ID	Order_Date	Status	Total_Amount
O001	C001	2024-02-10 09:30:00	Delivered	120.50
O002	C002	2024-03-02 10:15:00	Delivered	420.00
O003	C003	2024-03-03 12:45:00	Delivered	195.00
O004	C004	2024-03-05 08:25:00	Pending	230.00
O005	C005	2024-03-06 14:10:00	In Transit	365.00

Figure 7: Order Table

4.3.3.5 Order_Item Table

Field Name	Key Type	Data Type	Size	Description
Order_Item_ID	Primary Key	VARCHAR	10	Unique identifier for each order item
Order_ID	Foreign Key	VARCHAR	10	Linked to Order Table
Product_ID	Foreign Key	VARCHAR	10	Linked to Product Table
Quantity		INT	11	Number of product units ordered
Subtotal		DECIMAL	(10,2)	Total price for the item (Quantity × Price)

Table 6: Order_Item Table Design

Order_Item_ID	Order_ID	Product_ID	Quantity	Subtotal
OI001	O001	P001	2	241.00
OI002	O002	P004	3	270.00
OI003	O002	P008	2	110.00
OI004	O003	P007	1	110.00
OI005	O003	P009	1	85.00
OI006	O004	P002	3	180.00
OI007	O004	P010	2	90.00
OI008	O005	P005	2	150.00
OI009	O005	P006	1	180.00
OI010	O005	P009	1	35.00

Figure 8: Order Item Table

4.3.3.6 Payment Table

Field Name	Key Type	Data Type	Size	Description
Payment_ID	Primary Key	VARCHAR	10	Unique identifier for payment transaction
Order_ID	Foreign Key	VARCHAR	10	Linked to Order Table
Payment_Method		VARCHAR	20	Method of payment (“Cash”, “Card”, “Mobile Payment”)
Payment_Status		VARCHAR	20	Status of payment (“Pending”, “Paid”, “Failed”)
Payment_Date		DATETIME	—	Date and time of transaction

Table 7: Payment Table Design

Order_Item_ID	Order_ID	Product_ID	Quantity	Subtotal
OI001	O001	P001	2	241.00
OI002	O002	P004	3	270.00
OI003	O002	P008	2	110.00
OI004	O003	P007	1	110.00
OI005	O003	P009	1	85.00
OI006	O004	P002	3	180.00
OI007	O004	P010	2	90.00
OI008	O005	P005	2	150.00
OI009	O005	P006	1	180.00
OI010	O005	P009	1	35.00

Figure 9: Order Table

4.3.3.7 Delivery Table

Field Name	Key Type	Data Type	Size	Description
Delivery_ID	Primary Key	VARCHAR	10	Unique identifier for each delivery
Order_ID	Foreign Key	VARCHAR	10	Linked to Order Table
Delivery_Status		VARCHAR	20	Status of the delivery (“Out for Delivery”, “Delivered”)
Delivery_Agent		VARCHAR	50	Name of assigned delivery personnel
Delivery_Date		DATETIME	—	Date and time of successful delivery

Table 8: Delivery Table Design

Delivery_ID	Order_ID	Delivery_Status	Delivery_Agent	Delivery_Date
D001	O001	Delivered	Aziz Khan	2024-02-10 12:00:00
D002	O002	Delivered	Hasan Ali	2024-03-02 15:00:00
D003	O003	Delivered	Tuhin Ahmed	2024-03-03 17:00:00
D004	O004	Scheduled	Rasel Hossain	2024-03-05 11:00:00
D005	O005	Out for Delivery	Rubel Khan	2024-03-06 16:00:00

Figure 10: Order Table

Notes:

- All primary keys are defined as **VARCHAR(10)** for consistency and scalability.
- Foreign key constraints ensure **referential integrity** across dependent tables.
- **DECIMAL(10,2)** is used for monetary values to maintain precision.
- **VARCHAR** types are chosen for alphanumeric fields such as IDs, names, and categorical data.
- **DATETIME** fields record important time-based events like order placement and delivery completion.